



BUILDING STATEFUL AGENTIC AI WITH LANGGRAPH AND LLAMA 3

Project Overview

In this project, you will move beyond simple, stateless LLM wrappers to build a robust, production-ready **Stateful Agentic AI System**. You will act as an AI Engineer, leveraging the power of **LangGraph** to create cyclical, memory-aware workflows that solve complex problems.

You will implement three distinct solutions within a single modular application: a **Basic Chatbot**, a **Tool-Augmented Agent** (capable of writing and executing code, like the Snake Game example), and an **AI News Aggregator**.

The core of this project focuses on **End-to-End Implementation**. You will not just write a script; you will architect a professional folder structure, set up virtual environments, and implement "Modular Coding" practices used in top tech companies. Finally, you will deploy your application to the cloud using **Hugging Face Spaces**, making your portfolio live and accessible to the world.

What you will Learn?

- **LangGraph Fundamentals:** Master the creation of "Stateful Graphs." Unlike standard chains, these graphs maintain memory ("state") across user interactions, allowing for deeper, context-aware conversations.
- **Tool Usage & Reasoning:** Build agents that don't just talk but *do*. You will program your agent to understand when to use external tools—demonstrated by generating functional Python code for

a Snake game upon request.

- **High-Performance LLM Inference:** Integrate **Groq** to run open-source models like **Llama 3** with blazing-fast inference speeds, ensuring your chatbot feels instant and responsive.
- **Modular Coding Architecture:** Learn industry-standard software engineering. You will move away from messy notebooks to writing clean, reusable, and modular Python code.
- **Cloud Deployment:** Complete the lifecycle by deploying your Streamlit application to **Hugging Face Spaces**, giving you a shareable link for your resume.

Why this Project?

- **Stateful Intelligence:** Most beginner projects are "stateless" (they forget what you said 5 seconds ago). This project teaches you **State Management**, a critical skill for building real AI assistants.
- **Real-World "Agentic" Behavior:** You aren't just calling an API; you are building an agent that can act on your behalf, like generating and explaining code dynamically.
- **Professional Deployment:** You walk away with a live URL on Hugging Face, proving to recruiters that you can ship software, not just train models.

Module

Module 1: Project Initialization & Setup

1.0-Introduction.mp4

2.0- VS Code Project Set up.mp4

3- Set up git and github.mp4

Setting Up The Project Structure.mp4

Module 2: Interface & Model Configuration

5-Designing Streamlit Front End.mp4

6- Implementing the LLM Module.mp4

Module 3: Building the Agentic Graph

7-Implementing the Graph Builder Module.mp4

8-Implementing Node Functionalities.mp4

Module 4: Integration & Execution

9-Implementing the entire pipeline.mp4

10- Testing The End To End Application.mp4

Tech Stack & Tools

1. **Orchestration:** LangGraph (Stateful Agentic Workflows).
2. **LLM Provider:** Groq (High-speed inference).
3. **Model:** Llama 3 (Open-source LLM).
4. **Frontend:** Streamlit (Interactive Chat UI).

5. **Deployment:** Hugging Face Spaces.
6. **Language:** Python (Modular coding standards).